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Self-driving car tunnel

Elon Musk's The Boring Company is almost done with its first operational "loop tunnel" in Las Vegas. <http://digit.in/oct20-78>

Hold off on those new RTX 3000 cards

High-end graphics cards shouldn't be conking off this frequently

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he recent NVIDIA launch was shrouded in much secrecy. Given the plethora of rumours

that came out much before the launch, you'd think there was hardly anything held secret. Us consumers got the usual amount of information prior to launch but NVIDIA's card partners allegedly got next to no information about the cards prowess. Certain blogs with inside sources at partner companies have said that the partners were given just a barebones driver that could only power up the graphics card and allow them to test for temperatures and stability. We can understand such level of secrecy around a product category that's about to get very competitive but when your partners have to redesign certain elements of the graphics cards, then a barebones driver would put them at a significant disadvantage.

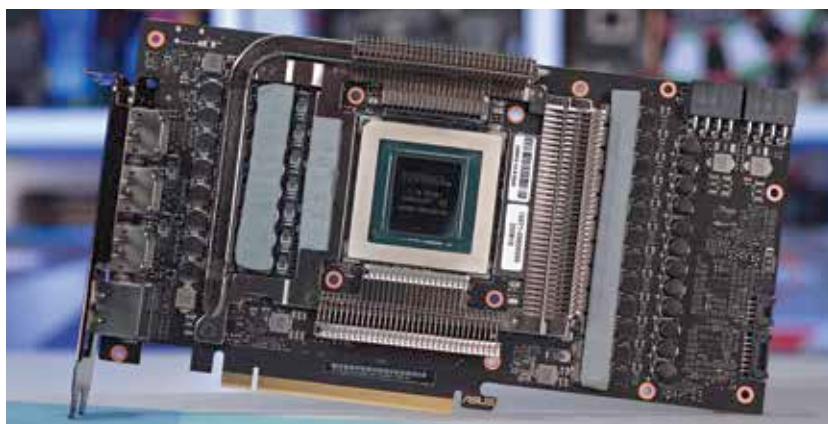
About a week after the launch of the RTX 3000 cards, reports of consumers who had purchased NVIDIA RTX 3080 partner cards facing stability issues started pouring in. There were enough complaints that NVIDIA partners had to issue official statements. That should give you an indication into the seriousness of the matter. The partners are still investigating the actual reason

for the crashes and some folks have already gone into detail as to why they believe the cards are crashing during gaming.

Igor, from Igor's Lab.de, has indicated that the capacitors are at fault. The graphics cards have several voltage control mechanisms which rely on sensors embedded in

frequency shifts and the voltage entering the GPU ends up switching just as rapidly.

The thing about electrical circuits is that they don't like such rapid switching action and would prefer a clean and smooth flow of current. Which is why we use capacitors to smoothen out erratic ripples in



NVIDIA RTX 3080 capacitor issues don't bode well

the GPU and across the PCB. These little things perform near-real-time adjustments after checking all the sensors within the GPU and the PCB. These sensors notice rapid switching of voltage levels, when we say rapid, we're talking 3-digit KHz and if we're to take an example, then 300 KHz literally means 300,000 variations per second. The voltage regulator mechanisms are fast enough to respond to such high-

current flow. Capacitors charge and discharge themselves, so there's a bit of switching happening within capacitors as well, it's just that the output comes out smoother than what went into the capacitor. Now, what happens if the capacitor isn't fast enough when it comes to switching? Sometimes, the current flow drops a little too much. Enough to cause the GPU to shut down for a fraction and whatever application is